

Production of Recombinant Human Antithrombin by *Pichia pastoris*

SHINOBU KUWAE,^{1*} MASAO OHYAMA,¹ TOMOSHI OHYA,¹
HIDEYUKI OHI,¹ AND KAORU KOBAYASHI¹

*Protein Research Laboratory, Pharmaceutical Research Division, Mitsubishi Pharma Corporation,
2-25-1 Shodai-ohsaka, Hirakata, Osaka 573-1153, Japan¹*

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This paper deals with the production of recombinant human antithrombin (rAT) by the methylotrophic yeast *Pichia pastoris*. In preliminary methanol-limited fed-batch fermentation, the rAT concentration reached 324 mg/l at 192 h of cultivation, but the specific heparin cofactor (HC) activity of rAT in the culture supernatant was 10% of that of plasma-derived antithrombin (pAT). To improve the specific HC activity of rAT, effort was first focused on the optimization of culture pH and media composition, resulting in protection of rAT against pH-dependent instability and proteolytic degradation. However, even in the optimized methanol-limited fed-batch fermentation, the specific HC activity of rAT in the culture supernatant was still 20% that of pAT. To investigate the unknown mechanisms involved in the decreased specific HC activity of rAT, the culture supernatant of mock-transfected cells was prepared by methanol-limited fed-batch fermentation. When pAT was added to this supernatant, a rapid decrease in HC activity was observed; the residual HC activity was 26% after 24 h of incubation at 25°C. The loss of pAT activity was prevented by addition of a formaldehyde scavenger, aminourea, to the supernatant. In addition, alcohol oxidase activity was observed in the supernatant, resulting in the accumulation of formaldehyde in the culture broth. These results suggest that the formaldehyde produced by methanol oxidation in the culture broth of *P. pastoris* might decrease the HC activity of rAT during fermentation. Replacing the methanol with glycerol as the carbon source improved the specific HC activity of rAT from 20% to above 40% of that of pAT. In the glycerol-limited fed-batch fermentation, rAT is expressed at 100 mg/l under the control of the truncated mutated *AOX2* promoter.

[Key words: *Pichia pastoris*, antithrombin, heparin cofactor activity, formaldehyde, methanol, glycerol]

Antithrombin (AT) is the major physiological serine proteinase inhibitor that inhibits several proteinases in the blood coagulation cascade, primarily thrombin and factor Xa, and therefore plays a crucial role in the regulation of blood coagulation (1, 2). Human AT is a single-chain glycoprotein with a molecular weight of 58 kDa, and contains 432 amino acids, three disulfide bridges and four carbohydrate side chains at Asn residues 96, 135, 155, and 192, which account for 15% of the total mass (3, 4). AT is normally present in plasma at a level of 150 mg/l (5). AT inactivates target proteinase by forming a 1:1 stoichiometric stable covalent complex with the reactive center Arg-393 of AT and reactive-site serine residue of the proteinase (6). The inhibitory activity of AT for thrombin is accelerated 1000-fold by the binding of heparin to AT (designated as heparin cofactor activity).

AT pharmaceutical products, derived from plasma, are used in therapy to manage acute thrombotic episodes and for prophylaxis during surgery and pregnancy in patients with AT deficiency. These products have been produced by

fractionation of plasma obtained from blood donors. However, varying blood source causes the potential risk of contamination of the product by blood-derived pathogens. Thus, the development of an alternative method of industrial preparation of AT is desirable. The production of human AT by recombinant DNA technology would address concerns of contamination with blood-derived pathogens. The methylotrophic yeast *Pichia pastoris* has been widely used as an efficient host for the production of heterologous proteins (7). The success of the *Pichia* expression system is attributed firstly to the use of a strong, tightly regulated alcohol oxidase (*AOX1*) promoter. In addition, the yeast is capable of growing to high cell densities, in excess of 130 g-DCW/l, in very simple and inexpensive mineral media using methanol as the sole carbon source (8). The strong promoter combined with high-cell-density fermentation has enabled production of recombinant proteins at extremely high intracellular (9, 10) and extracellular levels (11, 12).

We have previously reported the characterization of recombinant human antithrombin (rAT) expressed by the methylotrophic yeast *P. pastoris* (13). rAT was secreted into the medium and purified to over 99% by a combination of chromatography including heparin-affinity chromatography.

* Corresponding author. e-mail: Kuwae.Shinobu@mc.m-pharma.co.jp
phone: +81-(0)72-856-9295 fax: +81-(0)72-864-2341

Whereas the heparin-binding affinity of purified rAT was 10-fold higher than that of plasma-derived AT (pAT), the heparin cofactor (HC) activity was half that of pAT. This decreased HC activity was attributed to the partial *O*-linked glycosylation at Thr-386 in the reactive center loop of rAT.

In this paper, we report the other mechanisms that cause decreased HC activity of rAT during cultivation and the subsequent development of a fermentation process to improve the specific HC activity of rAT. We observed that the specific HC activity of rAT decreased with increasing cultivation time to 10% that of pAT, indicating that the loss of HC activity occurred during cultivation. In methanol-limited fed-batch fermentation, the specific HC activity of rAT was improved from 10% to 20% that of pAT by optimizing the culture pH and media composition, which protected rAT against pH-dependent instability and proteolytic degradation. In addition, we found that formaldehyde present in the culture supernatant could inactivate the HC activity of rAT. To avoid formaldehyde formation, methanol was replaced with glycerol as the carbon source, which improved the specific HC activity of rAT in the culture supernatant to above 40% that of pAT.

MATERIALS AND METHODS

Plasma-derived antithrombin (pAT) The preparation used in the experiments was the commercially available pAT Neuart® (Mitsubishi Pharma Corporation, Osaka).

Yeast strain *P. pastoris* GTS115 (*his4*), which was provided by the Phillips Petroleum Company (Bartlesville, OK, USA) and identical to strain GS115 (14), was used as the host strain for yeast transformation.

Plasmids An AT-expression plasmid pAT101 was described previously (13). To construct another antithrombin-expression plasmid with stronger promoter activity, the 1.4-kb *AsuII/BamHI* fragment containing the AT coding region fused with the yeast invertase (*SUC2*) pre-peptide was obtained from pAT101 and ligated to the 6.5-kb *AsuII/BamHI* fragment derived from the *P. pastoris* human serum albumin expression plasmid pHO068FRR (15), resulting in the AT-expression plasmid pIM102. Both AT-expression plasmids carried the mature AT cDNA region placed under control of the distinct truncated mutated *AOX2* promoters. The promoter of plasmid pAT101 contains a single copy of the native *AOX2*-UAS and a mutated *AOX2*-URS2, whereas the promoter of plasmid pIM102 carries triple copies of the synthetic *AOX2*-UAS but no *AOX2*-URS1 or *AOX2*-URS2. To construct a mock plasmid, the 7.7-kb *AsuII/BamHI* fragment was obtained from the *P. pastoris* human serum albumin-expression plasmid pMM042 (15), blunted, and self-ligated, yielding the mock plasmid pMM043.

Yeast transformation Each of the AT-expression and mock plasmids was linearized by digestion with *StuI*, a site within the *HIS4* gene, for integration into the *his4* locus on the chromosome of *P. pastoris* GTS115 (*his4*). The GTS115 strain was transformed with the linearized plasmid using the lithium acetate method (16) with slight modification for *P. pastoris*. The RH101 strain, which was a single-copy transformant with plasmid pAT101, was described previously (13). A double-copy transformant with the AT-expression plasmid pIM102 was chosen as another AT-producing strain by genomic Southern blot analysis and named the IM102 strain. A double-copy transformant with the mock plasmid was selected as mock-transfected cells.

Shake flask culture For the preparation of a seed culture, a single colony of each strain was inoculated from the stock plate

into an Erlenmeyer flask containing 50 ml of YPD medium (1% yeast extract, 2% peptone, 2% dextrose) and the culture was incubated at 30°C for 24 h with shaking at 120 rpm.

To investigate the effect of formaldehyde scavengers on cell growth, each formaldehyde scavenger was added to YPM medium (1% yeast extract, 2% peptone, 2% methanol) at 50 mM and the pH was adjusted to between 6.5 and 6.6. One ml of the seed culture was inoculated into Erlenmeyer flasks containing 50 ml of one of the media. The flasks were incubated at 30°C for 50 h with shaking at 120 rpm.

Culture samples were taken once a day for analysis. Cell growth was monitored using a model UV-1200 spectrophotometer (Shimadzu, Kyoto) at a wavelength of 540 nm.

Methanol-limited fed-batch fermentation Cells were pre-cultivated in YPD medium at 30°C for 24 h. Fed-batch fermentation was conducted in a 10-l fermentor (Bio Master D type; Able, Tokyo) which contained 2.66 l of basal medium A. Basal medium A consisted of the following (per liter of deionized water): 50 g glycerol, 21 ml H_3PO_4 (85%), 0.9 g $CaSO_4 \cdot 2H_2O$, 14.3 g K_2SO_4 , 11.7 g $MgSO_4 \cdot 7H_2O$, 3.9 g KOH (85%), 50 g peptone, 91 mg $FeSO_4 \cdot H_2O$, 8.4 mg $CuSO_4 \cdot 5H_2O$, 5.74 mg $MnSO_4 \cdot 4-5H_2O$, 28 mg $ZnSO_4 \cdot 7H_2O$, 12.9 μ l H_2SO_4 , 1.12 mg KI, 0.28 mg $Na_2MoO_4 \cdot 2H_2O$, 0.028 mg H_3BO_3 , 0.378 mg $CoCl_2$, 0.12 mg D-biotin, 0.84 mg thiamine hydrochloride, and 0.3 g Adekanol LG109. Concentrations of H_3PO_4 , K_2SO_4 , KOH, $MgSO_4$, and $CaSO_4$ in basal medium A were increased when needed by 20% (basal medium B) or 50% (basal medium C).

The cells were grown batchwise until the initial glycerol was depleted (~24 h). When the glycerol in the basal medium was depleted, as indicated by a dissolved oxygen spike, continuous feeding of methanol-fed batch medium and 32% peptone was started. The methanol-fed batch medium contained (per liter of methanol) the following: 0.325 g $FeSO_4 \cdot H_2O$, 0.03 g $CuSO_4 \cdot 5H_2O$, 20.5 mg $MnSO_4 \cdot 4-5H_2O$, 0.1 g $ZnSO_4 \cdot 7H_2O$, 25 μ l H_2SO_4 , 4 mg KI, 1 mg $Na_2MoO_4 \cdot 2H_2O$, 0.1 mg H_3BO_3 , 0.135 mg $CoCl_2$, 0.8 mg D-biotin, and 12 mg thiamine hydrochloride. The feed rate of the methanol-fed batch medium was controlled by feedforward control mode using the following equations, and cells were grown under methanol limitation.

$$F(t) = 5.29 \exp(0.02 \times t) \quad (0 \leq t \leq 77.1) \quad (1)$$

$$F(t) = 24.7 \quad (t \geq 77.1) \quad (2)$$

where $F(t)$ is the feed rate (g/h) of the methanol-fed batch medium at time t after the feeding started (h). The feed rate of 32% peptone was 0.26 g per gram feeding of the methanol-fed batch medium. During batch mode, the fermentor was operated at a temperature of 30°C and the pH was maintained at 5.85 by addition of 28% ammonia solution. After feeding began, the temperature was shifted from 30°C to 25°C and the pH was changed when needed. The agitation rate was increased to maintain the level of dissolved oxygen above 10% of air saturation.

Culture samples were taken once a day for analysis. Cell growth was monitored using a model UV-1200 spectrophotometer (Shimadzu) at a wavelength of 540 nm. The supernatant was obtained by centrifugation at 10,000 rpm for 5 min and stored at -80°C until analysis. A cell-free culture supernatant of mock-transfected cells was prepared by filtration of the supernatant through a 0.22- μ m filter (Millex GV; Millipore, Billerica, MA, USA) and stored at -80°C until use.

Glycerol/methanol mixed-feed and glycerol-feed fed-batch fermentation Cells were pre-cultivated in YPD at 30°C for 24 h. Mixed-feed and glycerol-feed fed-batch fermentation was conducted in a 3-l fermentor (Bio Master D type; Able), which contained 0.84 l of basal medium C. The mixed-fed batch medium consisted of: 742.5 g glycerol, 7.5 g methanol, 250 g deionized water, 0.305 g $FeSO_4 \cdot H_2O$, 0.0281 g $CuSO_4 \cdot 5H_2O$, 19.2 mg

MnSO₄·4H₂O, 93.7 mg ZnSO₄·7H₂O, 23.4 mg H₂SO₄, 3.74 mg KI, 0.94 mg Na₂MoO₄·2H₂O, 0.094 mg H₃BO₃, 0.127 mg CoCl₂, 0.75 mg D-biotin, and 11.7 mg thiamine hydrochloride. The glycerol-fed batch medium was the same as the mixed-fed batch medium except that it consisted of 750 g glycerol as the sole carbon source. After depletion of the initial glycerol in the basal medium, continuous feeding of each fed batch medium and 32% peptone was started. The feed rate of the fed batch medium was controlled by feedforward control mode using the following equations, and cells were grown under carbon limitation:

$$F(t) = 2.97 \exp(0.03 \times t) \quad (0 \leq t \leq 41.6) \quad (3)$$

$$F(t) = 10.35 \quad (t \geq 41.6) \quad (4)$$

where $F(t)$ is the feed rate (g/h) of the fed batch medium at time t after the feeding started (h). The feed rate of 32% peptone was 0.196 g per gram feeding of the fed batch medium. As a positive control of the mixed-fed fed batch fermentation, methanol-fed batch medium was fed at the same rate as the mixed-fed batch medium on the weight basis of the carbon source; namely, methanol-fed batch medium was fed at 75% of the feed rate of the mixed-fed batch medium. During batch mode, the fermentor was operated at a temperature of 30°C and the pH was maintained at 5.85 by addition of 28% ammonia solution. After feeding began, the temperature was shifted from 30°C to 25°C and the pH was adjusted from 5.85 to 6.5. The agitation rate was increased to maintain the level of dissolved oxygen above 10% of air saturation.

Antithrombin determination and heparin cofactor activity assay The concentration of AT in the culture supernatant was determined by ELISA. The ELISA made use of a rabbit anti-human AT polyclonal antibody (DakoCytomation, Copenhagen, Denmark) and sheep anti-human AT affinity-purified IgG conjugated to horseradish peroxidase (Cedarlane, Hornby, Ontario, Canada).

Antithrombin activity was measured as HC activity with a commercial assay kit that uses the thrombin-specific chromogenic substrate S-2238 (TESTZYM ATIII 2 kit; Daiichi Pure Chemical, Tokyo). The test sample was diluted 2–20-fold with 100 mM 2-amino-2-hydroxymethyl-1,3-propanediol containing metylamine hydrochloride, pH 7.4, to give a final heparin cofactor activity of 25 IU/l or less. The diluted sample (25 µl) was added to a 100-µl mixture of thrombin and heparin for a 5-min preincubation at 37°C before the initiation of the reaction by adding 100 µl of S-2238 (1.43 g/l). The reaction was stopped after a period of 5 min by the addition of citric acid solution (1 ml) and then the absorbance at 405 nm was determined.

Specific heparin cofactor activity was calculated by dividing volumetric heparin cofactor activity by the AT concentration determined by an ELISA specific for AT.

SDS-PAGE and Western blot analysis The culture supernatant was diluted 1:10 with deionized water, and 50 µl of the diluted supernatant was mixed with 25 µl of NuPAGE LDS sample buffer (Invitrogen, Carlsbad, CA, USA), 15 µl of deionized water, and 10 µl of NuPAGE sample reducing agent (Invitrogen). The mixture was incubated at 70°C for 10 min and then 10 µl was subjected to electrophoresis. SDS-PAGE was performed using NuPAGE 4–12% Bis-Tris Gel in MOPS SDS running buffer under the electrophoresis conditions given by the manufacturer (Invitrogen). Proteins separated by SDS-PAGE were electrophoretically transferred to a poly(vinylidene difluoride) (PVDF) membrane (Tefco, Tokyo). The membrane was incubated in blocking buffer (5% skim milk, PBS) for at least 1 h and then incubated with 1:1000 diluted polyclonal antibody raised in rabbits against human AT (Oriental Yeast, Tokyo) for 1 h at room temperature. After being washed with PBS containing 0.1% Tween₂₀, the membrane was incubated with 1:1000 diluted horseradish peroxidase-conjugated goat anti-rabbit IgG (H+L) antibody (Bio-Rad, Hercules, CA,

USA) for 30 min at room temperature. The membrane was washed with 20 mM Tris (pH 7.5)/0.5 M NaCl and then incubated in a chromogenic substrate solution (POD Immunostain set; Wako Pure Chemical, Osaka).

Analysis of formaldehyde in the culture supernatant

Formaldehyde was measured by using Formaldehyde-Test Wako (Wako Pure Chemical). The formaldehyde measuring kit made use of the reaction between formaldehyde and 4-amino-3-hydrazino-5-mercapto-1,2,4-triazole at a 1:1 molar ratio under alkaline conditions, and the reaction product, 6-mercapto-S-triazolo[4,3-b]-s-tetrazine, was measured at 550 nm.

RESULTS AND DISCUSSION

rAT production on methanol The rAT-producing recombinant *Pichia* strain RH101, which was a single-copy transformant with the plasmid pAT101, was cultivated in basal medium A. The culture pH was maintained at pH 5.85 by adding 28% ammonia solution during the course of cultivation. After depletion of glycerol in the culture (23 h), both methanol-fed batch medium and 32% peptone were continuously fed under methanol-limited mode as described in Materials and Methods. The concentration of rAT in the culture supernatant was determined by an ELISA specific for AT. Specific HC activity of rAT was calculated by dividing volumetric HC activity by the AT concentration. The rAT concentration increased with time for 192 h and reached 324 mg/l (Fig. 1A), whereas the specific HC activity of rAT in the culture supernatant decreased gradually with time to

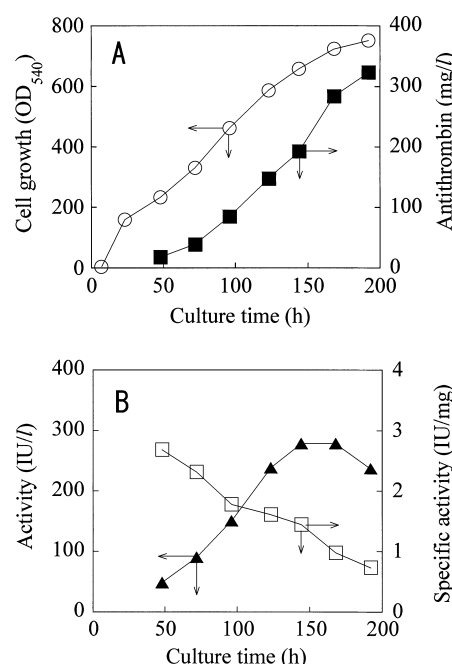


FIG. 1. rAT production in methanol-limited fed-batch fermentation with *P. pastoris* strain RH101. The culture was grown in glycerol batch mode for 0–23 h, after which rAT expression was induced by initiating a methanol-fed batch at pH 5.85. (A) Cell concentration (open circles) was expressed as OD₅₄₀. rAT concentration (closed squares) in the culture supernatant was determined by an ELISA specific for antithrombin. (B) Specific HC activity of rAT (open squares) was calculated by dividing HC activity (closed triangles) by the amount of rAT in the culture supernatant.

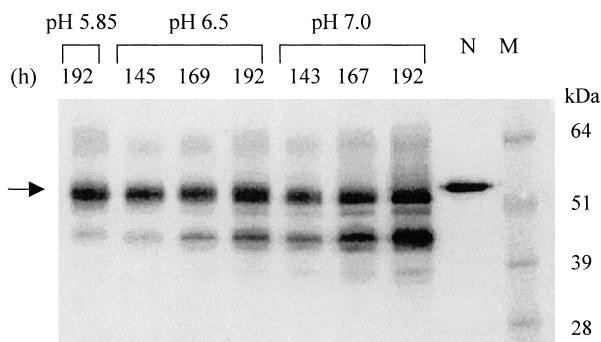


FIG. 2. Western blot analysis of rAT in culture supernatant from fed-batch fermentations at pH 5.85, 6.5, and 7.0. Lane 1, 0.5 μ l of supernatant from fed-batch fermentation at pH 5.85, collected at 192 h; lanes 2–4, 0.5 μ l of supernatant from fed-batch fermentation at pH 6.5, collected at 145, 169, and 192 h, respectively; lanes 5–7, 0.5 μ l of supernatant from fed-batch fermentation at pH 7.0, collected at 143, 167, and 192 h, respectively; lane 8, purified plasma-derived human antithrombin; lane 9, prestained molecular weight markers (Seeblue; Invitrogen). The position of intact rAT is indicated with an arrow.

0.73 IU/mg (Fig. 1B). This level of specific activity corresponds to only 10% of that of plasma-derived antithrombin (pAT). In the *Pichia* expression system, reduced activity of secreted recombinant proteins during cultivation is often caused by proteolytic degradation of the proteins. To investigate whether the decreased specific HC activity was caused by proteolytic degradation, Western blot analysis of the culture supernatant was performed. Little evidence of proteolytic degradation of rAT was observed at 192 h of cultivation at a culture pH of 5.85 (Fig. 2). This result indicated that the reduced HC activity of rAT during cultivation was not caused by degradation of rAT. Preliminary experiments on the pH stability of pAT in buffer solutions showed that pAT was unstable at acidic pH and was stable at neutral pH (data not shown). To evaluate whether specific HC activity of rAT is improved at higher culture pH than pH 5.85, the rAT-producing cells were cultivated at pH values of 6.5 and 7.0 using basal medium A. As shown in Table 1, the maximum cell concentration decreased with increasing culture pH from 5.85 to 7.0, indicating that the optimal pH for cell growth is below pH 6.5. The highest rAT concentration (324 mg/l) was observed at a culture pH of 5.85, but the highest HC activity (290 IU/l) and the highest specific activity (1.13 IU/mg) was observed at a culture pH of 6.5. Western blot analysis of the culture supernatants showed that the

amount of degraded rAT fragments increased with increasing culture pH from 5.85 to 7.0 (Fig. 2). Although the least amount of degraded rAT fragments was observed at pH 5.85, the specific activity of rAT was lowest among the culture pHs tested. This result suggests that the reduced specific HC activity at a culture pH of 5.85 was at least partially caused by pH-dependent instability of rAT. To avoid the instability of rAT at pH 5.85, we selected cultivation at pH 6.5 for the production of rAT and then tried to minimize the proteolytic degradation at pH 6.5.

Several strategies have been reported as effective in minimizing the proteolytic instability of specific foreign proteins secreted into the *P. pastoris* culture medium. The first is the addition to the culture medium of amino acid-rich supplements such as peptone or casamino acids, which reduce product degradation possibly by acting as excess substrates for one or more problem proteases (12, 17). In the present work, a basal medium containing 5% peptone was used, and 32% peptone was continuously fed to keep the final concentration above 5%. Despite the addition of peptone to the medium, degradation of rAT was observed at pH values above 6.5. The second strategy to minimize proteolytic degradation is to optimize the culture pH between pH 3.0 and pH 7.0 (12, 18, 19). Although proteolytic degradation of rAT was minimized at pH 5.85 as described above, the specific activity at pH 5.85 was lower than that at pH 6.5 probably due to pH-dependent instability of rAT. The third strategy involves elevating the level of ammonium in the culture broth adequately, which is explained by the hypothesis that protease activity is induced under nitrogen starvation (19). To test this strategy, the level of ammonium was elevated by increasing the H_3PO_4 concentration because the culture pH was adjusted to 6.5 by the addition of a 28% ammonia solution. For this purpose, the basal media B and C were formulated so that concentrations of H_3PO_4 , K_2SO_4 , KOH, $MgSO_4$, and $CaSO_4$ were increased by 20% (basal medium B) and 50% (basal medium C) in comparison with those of basal medium A. There was little difference in the amount of 28% ammonia solution consumed during cultivation under the three conditions (data not shown). As shown in Table 1, the specific HC activity of rAT was improved from 1.13 to 1.48 IU/mg by using basal medium C. The amount of degraded rAT fragments decreased by increasing initial salt concentrations (data not shown). A possible further increase in the ammonium concentration in the culture was examined by feeding a solution of $(NH_4)_2HPO_4$ during cultivation, but

TABLE 1. Effect of culture pH and basal medium on rAT production and specific heparin cofactor activity^a

Culture pH	Basal medium ^b	Maximum cell concentration (OD ₅₄₀)	Antithrombin concentration ^c (mg/l)	Heparin cofactor activity ^c (IU/l)	Specific heparin cofactor activity ^c (IU/mg)
5.85	A	752	324	238	0.73
6.5	A	707	257	290	1.13
7.0	A	617	264	231	0.88
6.5	A	707	257	290	1.13
6.5	B	688	250	356	1.42
6.5	C	678	263	387	1.48

^a Methanol-limited fed-batch fermentations with *P. pastoris* strain RH101 were conducted at pH 5.85, 6.5, and 7.0 using basal media A, B, and C.

^b Basal media B and C contain 20% and 50%, respectively, increased concentrations of H_3PO_4 , K_2SO_4 , KOH, $MgSO_4$, and $CaSO_4$ in comparison with those of basal medium A.

^c rAT concentrations, heparin cofactor activities, and specific heparin cofactor activities were determined at 192 h of cultivation.

further improvement of the specific activity was not achieved (data not shown).

Thus, the optimized culture conditions, including culture pH and medium composition, protected rAT against pH-dependent instability and proteolytic degradation and improved the specific HC activity of rAT from 0.73 to 1.48 IU/mg. However, the specific HC activity of rAT is still 20% of that of pAT. These results led us to further investigate causal factors responsible for the decreased specific HC activity of rAT in the culture supernatant.

Formaldehyde inactivates HC activity of pAT To determine whether the decrease in HC activity of rAT occurred extracellularly, culture supernatants from mock-transfected *Pichia* cells were prepared. The methanol-limited fed-batch fermentation of the mock-transfected cells was performed using basal medium C at pH 6.5. The supernatants of mock-transfected cells were collected at 48 h and again at 144 h of cultivation and then filtered through a 0.22- μ m filter to completely remove the cells from the supernatant (Mock supernatant). Plasma-derived AT (pAT) with a specific HC activity of 7.0 IU/mg was added to Mock supernatant collected at 144 h of cultivation to give a final concentration of 500 IU/l. The initial HC activity of 530 IU/l was determined immediately after the addition of pAT to Mock supernatant, indicating Mock supernatant did not interfere with the assay of HC activity. Then, the time course of the residual activity was measured after 2, 7, and 24 h of incubation at 25°C. As shown in Fig. 3, a rapid decrease in HC activity was observed after 2 h of incubation; the residual HC activities were 41% (2 h), 33% (7 h), and 26% (24 h). This observation suggests the presence of a highly reactive substance that could inactivate pAT in Mock supernatant. Furthermore, adding methanol to Mock supernatant at 1 g/l accelerated the rate of decline in HC activity of pAT (Fig. 3). Interestingly, 16% of HC activity still remained after 24 h of incubation. Methanol at a concentration of 1 g/l in a potassium phosphate buffer (pH 6.5) did not inactivate the HC

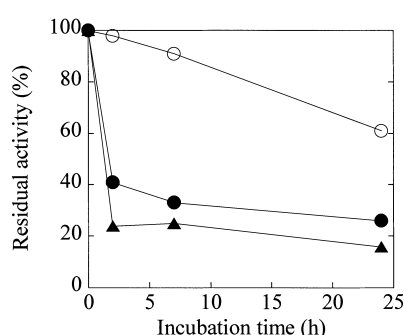


FIG. 3. Inactivation of HC activity of pAT in the culture supernatant from methanol-limited fed-batch culture of mock-transfected *P. pastoris*. Methanol-limited fed-batch fermentation with mock-transfected *P. pastoris* was performed using basal medium C at pH 6.5. The culture supernatant was collected at 144 h of cultivation and then filtered through a 0.22- μ m filter to completely remove the cells. pAT was added to a final concentration of 500 IU/l to the filtered supernatant and incubated at 25°C with and without the addition of methanol and aminourea. Symbols: closed circles, without addition of methanol and aminourea; closed triangles, with addition of methanol at 1 g/l; open circles, with addition of methanol and aminourea at 1 g/l and 50 mM, respectively.

activity of pAT (data not shown). Therefore, we hypothesized that formaldehyde, which could be produced by methanol oxidation, might inactivate pAT in Mock supernatant. To test the hypothesis, a formaldehyde scavenger, aminourea (20), was added to Mock supernatant. As shown in Fig. 3, the addition of aminourea to Mock supernatant at a final concentration of 50 mM prevented the loss of HC activity. This result indicates that the loss of HC activity of pAT in Mock supernatant was attributed in large part to an inactivation by formaldehyde and suggests that in a similar fashion the decreased HC activity of rAT produced by *P. pastoris* might be ascribed to an inactivation of rAT by formaldehyde.

The concentration of formaldehyde in culture supernatants was determined spectrophotometrically at 550 nm, based on a reaction of formaldehyde with 4-amino-3-hydrazino-5-mercapto-1,2,4-triazole. Formaldehyde was detected in Mock supernatant collected at 48 h and 144 h of cultivation at 3.8 mg/l and 10.8 mg/l, respectively. Formaldehyde was also observed in the culture supernatant from the rAT-producing strain (data not shown). The result that addition of methanol to Mock supernatant accelerated the declining rate of HC activity suggests the presence of alcohol oxidase activity in Mock supernatant. To confirm the presence of alcohol oxidase activity in Mock supernatant, production of formaldehyde and hydrogen peroxide was examined by adding methanol to Mock supernatant. When methanol was added at 1 g/l to Mock supernatant collected at 48 h and 144 h of cultivation, formaldehyde was produced at 64 mg/l and 171 mg/l, respectively, after 1 h of incubation at 25°C. This result indicates that alcohol oxidase activity increased with increasing culture time in the culture supernatant of *P. pastoris* grown on methanol. The production of hydrogen peroxide in Mock supernatant was also detected by the formation of quinoneimine from 4-aminoantipyrine and *N*-ethyl-*N*-(3-sulfopropyl)-*m*-anisidine (data not shown). We have previously reported the purification of vacuolar enzyme carboxypeptidase Y from the culture supernatant of *P. pastoris* (21). This vacuolar enzyme is thought to be leaked out to the culture supernatant by cell lysis during fed-batch fermentation. Likewise, alcohol oxidase in peroxisomes is considered to be released by cell lysis during cultivation.

Screening of formaldehyde scavengers The addition of a formaldehyde scavenger to the culture medium was expected to prevent the loss of HC activity of rAT during cultivation. Thus, we screened a variety of amino compounds for their capacity to prevent inactivation of pAT by formaldehyde using Mock supernatant. Each amino compound was added at 50 mM to Mock supernatant, to which pAT and methanol were added at the same time at 500 IU/l and 1 g/l, respectively. Then, the time course of residual HC activity was measured at 0, 2, 7 and 24 h of incubation at 25°C. Amino acids, including Lys, Arg, His and Asn, were found to have less potential than aminourea to prevent the loss of HC activity of pAT (data not shown). Among the amino compounds tested, *N,N'*-diaminourea and *N,N'*-diaminoguanidine showed higher capacity than aminourea (Fig. 4A). A lesser effect was found with aminoguanidine. Urea and guanidine had no detectable effect. Aminourea,

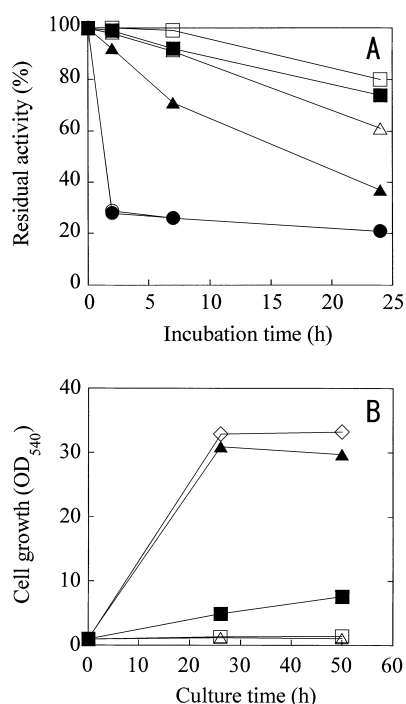


FIG. 4. (A) Screening of amino compounds for protection of HC activity. Each amino compound was added at 50 mM to the culture supernatant from methanol-limited fed-batch fermentation with mock-transfected *P. pastoris*. Time course of residual HC activity was obtained after addition of methanol and pAT at 1 g/l and 500 IU/l, respectively. (B) Effect of amino compounds on cell growth. Each amino compound was added at 50 mM to YPM medium. Symbols: open triangles, aminourea; open squares, *N,N'*-diaminourea; closed triangles, amino guanidine; closed squares, *N,N'*-diaminoguanidine; open circles, urea; closed circles, guanidine; open diamonds, no additive.

N,N'-diaminourea, aminoguanidine and *N,N'*-diaminoguanidine were selected as formaldehyde scavengers.

The effect of formaldehyde scavengers on the cell growth of the recombinant *Pichia* strain RH101 was investigated using a shake-flask culture. Each formaldehyde scavenger was added to YPM medium at 50 mM. Cell growth was scarcely inhibited by aminoguanidine but strongly inhibited by *N,N'*-diaminoguanidine, aminourea and *N,N'*-diaminourea (Fig. 4B). No cell growth was observed in the presence of aminourea and *N,N'*-diaminourea. From these results and since formaldehyde is a key intermediate in methanol metabolism, we considered that adding these formaldehyde scavengers to the culture medium in order to protect rAT from formaldehyde is difficult when cells are grown on methanol as the carbon source.

Glycerol/methanol mixed-feed process and glycerol-feed process In an attempt to prevent the decrease in specific HC activity of rAT during cultivation, mixed feeding of glycerol and methanol at a ratio of 99:1 was examined. The mixed-feed protocol was intended to minimize formaldehyde accumulation by reducing the amount of methanol used during fed-batch fermentation. In addition, the methanol-feed process and the glycerol-feed process were performed as a positive and negative control, respectively. The rAT-producing strain RH101 was cultivated in basal medium C. After depletion of glycerol in the culture,

the cells were grown in carbon-limited fed-batch mode using methanol-, glycerol- and mixed-feed batch medium, respectively. For the ease of comparison, each fed batch medium was fed at the same rate on the weight basis of carbon sources. The final rAT concentration obtained by the mixed-feed process was 40% of that obtained by the methanol-feed process (Fig. 5B). As expected, the specific HC activity of the mixed-feed process was higher than that of the methanol-feed process, but it decreased to 2.5 IU/mg at 144 h of cultivation (Fig. 5D). Even though the amount of methanol used was reduced by 99% on the weight basis of the carbon source, the mixed-feed process did not completely prevent the decrease in specific HC activity. Loss of specific HC activity was successfully prevented by the glycerol-feed process, which maintained specific HC activity at more than 3.0 IU/mg during cultivation. This result is very important for the production of rAT because therapeutic proteins must be manufactured in a way that ensures consistency within and across batches. The glycerol feed process achieved a final rAT concentration of 50 mg/l, which corresponds to 20% of that by the methanol-feed process. It is considered that carbon-limited growth on glycerol caused full derepression of rAT expression which was under the control of the mutated *AOX2* promoter and then resulted in considerably high expression of rAT.

To increase rAT expression level on glycerol, the recombinant *Pichia* strain IM102, which was a double-copy transformant with the rAT-expressing plasmid pIM102, was grown in a glycerol-limited fed-batch mode. As shown in Fig. 6A, the cell concentration increased until 144 h of cultivation. The rAT concentration and HC activity plateaued after 101 h of cultivation at 100 mg/l and 300 IU/l, respectively. The specific HC activity of rAT in the culture supernatant stayed above 3 IU/mg (Fig. 6B). The final rAT concentration was increased twofold by changing the recombinant strain from a single-copy transformant with plasmid pAT101 to a double-copy transformant with plasmid pIM102. A gene dosage effect of the rAT-expression cassette is considered to be responsible for the twofold increase in rAT expression. The changes in structure of the mutated *AOX2* promoters for rAT expression presumably did not affect its promoter activity under carbon-limited growth on glycerol because the mutated *AOX2* promoters were activated not by an induction mechanism but only by a derepression mechanism. A further increase in rAT expression level is expected by replacing the mutated *AOX2* promoters with strong promoter that work under growth on glycerol such as the glyceraldehyde-3-phosphate dehydrogenase (*GAP*) promoter (22).

The specific HC activity of rAT was improved from 1.48 IU/mg to above 3 IU/mg by replacing methanol with glycerol as the sole carbon source. This result supports the hypothesis that the decreased HC activity of rAT produced on methanol may be attributed, at least in part, to an inactivation of rAT by formaldehyde. We found no reports on the identification of formaldehyde as responsible for the inactivation of recombinant protein in a culture of *P. pastoris* grown on methanol. Formaldehyde is bound to proteins at various levels called reversible, acid-labile and acid-resistant. Tome *et al.* reported that the ϵ -amino group of lysine

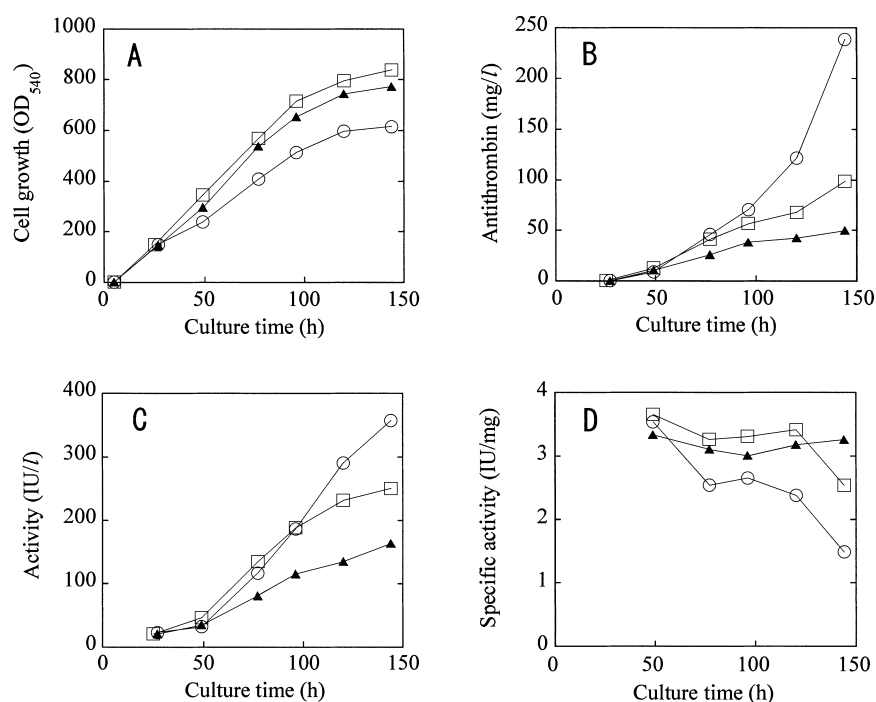


FIG. 5. Effect of carbon source on cell growth (A), rAT production (B), HC activity (C) and specific HC activity (D). Carbon-limited fed-batch fermentation with rAT-producing strain RH101 was performed using different carbon sources. Symbols: open circles, methanol-feed; open squares, mixed-feed of glycerol and methanol at ratio of 99:1; closed triangles, glycerol-feed.

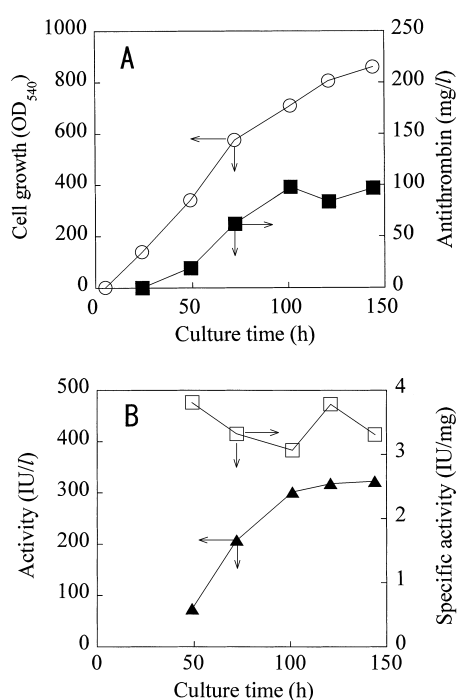


FIG. 6. rAT production in glycerol-limited fed-batch fermentation with *P. pastoris* strain IM102. The culture was grown in glycerol batch mode for 0–23 h, after which glycerol-limited fed-batch fermentation was carried out. (A) Cell concentration (open circles) was expressed as OD₅₄₀. rAT concentration (closed squares) in culture supernatant was determined by ELISA specific for AT. (B) Specific HC activity of rAT (open squares) was calculated by dividing heparin cofactor activity (closed circles) by the amount of rAT in the culture supernatant.

was largely involved in the combination of formaldehyde with bovine serum albumin at pH 7, forming reversible hydroxymethyl-lysine, an acid-labile lysine-arginine methylene bridge, and acid-resistant derivatives such as methyl-lysine, formyl-lysine, and a lysine-tyrosine methylene bridge (23). Gold *et al.* reported that the guanidyl group of arginine and the ϵ -amino group of lysine were hydroxymethylated in the reaction of formaldehyde with gelatin at pH 7.2 (24). In both cases, a lysine-arginine methylene bridge was observed as the major product. On the other hand, the x-ray crystal structure of an AT-heparin pentasaccharide complex has identified several basic residues of AT, Lys-11, Arg-13, Arg-46, Arg-47, Lys-114, Lys-125 and Arg-129, that participate in the heparin binding by interacting with negative groups of the heparin pentasaccharide (25). Substitution of Lys-114, Arg-129 and Lys-125 led to $\sim 10^5$ -, ~ 400 -, and ~ 400 -fold, respectively, losses of affinity for heparin pentasaccharide (26). The binding of heparin to AT is a requisite step in the activation of AT. Moreover, the study of natural antithrombin variants showed that substitution of Arg-393 in the reactive center of AT leads to impaired ability to interact with thrombin (27). Although the chemical structures of the reaction products of AT with formaldehyde remain unknown, the above-mentioned studies suggest that the lysine and arginine residues of AT may be modified with formaldehyde and this modification may lead to the decrease in HC activity of AT.

In a previous study, we reported that the specific HC activity of rAT purified from a culture on methanol was half that of pAT (13). This decreased specific HC activity was attributed to the partial *O*-linked glycosylation at Thr-386 in

the reactive center loop of rAT. In the present report, the specific HC activity of rAT produced on glycerol was improved to half that of pAT, so we expect that the specific HC activity of rAT from the culture on glycerol is improved by purification to above half that of pAT. Furthermore, it is expected that the glycerol-feed fed-batch fermentation process in combination with the replacement of Thr-386 in rAT with an appropriate amino acid to avoid *O*-glycosylation will lead to the production of rAT with specific HC activity comparable to that of pAT.

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